

Joshua Agterberg

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ACADEMIC EMPLOYMENT

2024-Present	University of Illinois Urbana-Champaign Assistant Professor Department of Statistics
2023-2024	University of Pennsylvania Postdoctoral Researcher Innovation in Data Engineering and Sciences (IDEAS) Initiative Department of Electrical and Systems Engineering Department of Statistics and Data Science Advised by Professors René Vidal and Yuxin Chen

EDUCATION

2017-2023	Johns Hopkins University
2019-2023	PhD in Applied Mathematics and Statistics Dissertation Title: <i>Asymptotics and Statistical Inference in High-Dimensional Low-Rank Matrix Models</i>
2017-2019	Master of Science in Engineering in Applied Mathematics and Statistics Advised by Professor Carey Priebe
2013-2017	University of Wisconsin-Madison Bachelor of Business Administration, Majors in Actuarial Science and Mathematics Advised by Professor Marjorie Rosenberg GPA: 3.73/4.0, Actuarial Science Major GPA: 4.0/4.0 Graduated with Distinction

RESEARCH INTERESTS

- Statistical Network Analysis
- High-dimensional Statistics
- Spectral Methods
- Mathematical Data Science
- Nonparametric Statistics
- Nonconvex Optimization
- Tensor Data Analysis
- Low-Rank Models

(* denotes students supervised or mentored)

1. **Joshua Agterberg**, Minh Tang, and Carey Priebe, “Nonparametric Two-Sample Hypothesis Testing for Low-Rank Random Graphs of Differing Sizes,” *Electronic Journal of Statistics*, 2026.
(*Won best presentation award in Nonparametric Statistics Student Competition, JSM 2021*)
2. **Joshua Agterberg** and Anru Zhang, “Statistical Inference for Low-Rank Tensors: Heteroskedasticity, Subgaussianity, and Applications,” *Annals of Statistics*, 2026+.
3. **Joshua Agterberg** and Joshua Cape, “An Overview of Asymptotic Normality in Stochastic Blockmodels: Cluster Analysis and Inference,” *Statistical Science*, 2026.
4. **Joshua Agterberg**, “Discussion of “Statistical Exploration of the Manifold Hypothesis” by Nick Whiteley, Annie Gray, and Patrick Rubin-Delanchy,” *Journal of the Royal Statistical Society, Series B*, 2026.
5. **Joshua Agterberg**, Zachary Lubberts, and Jesús Arroyo, “Joint Spectral Clustering in Multi-layer Degree-Corrected Stochastic Blockmodels,” *Journal of the American Statistical Association (Theory and Methods)*, 2025.
6. **Joshua Agterberg** and Anru Zhang, “Estimating Higher Order Mixed Memberships via the $\ell_{2,\infty}$ Tensor Perturbation Bound,” *Journal of the American Statistical Association (Theory and Methods)*, 2025.
7. Anton Alyakin, **Joshua Agterberg**, Hayden Helm, and Carey Priebe, “Correcting a Nonparametric Two-sample Graph Hypothesis Test for Graphs with Different Numbers of Vertices,” *Applied Network Science*, 2024.
8. Aranyak Acharyya, **Joshua Agterberg**, Michael Trosset, Youngser Park, and Carey Priebe, “Semi-supervised Regression in Latent Structure Networks on Unknown Manifolds,” *Applied Network Science*, 2023.
9. Zachary Pisano, **Joshua Agterberg**, Carey Priebe, and Daniel Naiman, “Spectral Graph Clustering via the Expectation-Solution Algorithm,” *Electronic Journal of Statistics*, 2022.
10. **Joshua Agterberg**, Zachary Lubberts, and Carey Priebe, “Entrywise Estimation of Singular Vectors of Low-Rank Matrices with Heteroskedasticity and Dependence,” *IEEE Transactions on Information Theory*, 2022.
11. Jaewon Chung, Bijan Varjavand, Jesús Arroyo, Anton Alyakin, **Joshua Agterberg**, Minh Tang, Joshua Vogelstein, and Carey Priebe, “Valid Two-Sample Graph Testing via Optimal Transport Procrustes and Multiscale Graph Correlation: Applications in Connectomics,” *Stat*, 2021.
12. **Joshua Agterberg**, Youngser Park, Jonathan Larson, Chris White, Carey Priebe, and Vince Lyzinski, “Vertex Nomination, Consistent Estimation, and Adversarial Modification,” *Electronic Journal of Statistics*, 2020.
13. Fanghao Zhong, Margie Rosenberg, **Joshua Agterberg**, and Richard Crabb, “Social Determinant-Based Profiles of US Adults with the Highest and Lowest Health Expenditures Using Clusters,” *North American Actuarial Journal*, 2020.
14. **Joshua Agterberg**, Fanghao Zhong, Richard Crabb, and Margie Rosenberg, “Cluster Analysis Application to Identify Groups of Individuals with High Health Expenditures,” *Health Services*

PREPRINTS

(* denotes students supervised or mentored)

1. Aranyak Acharyya, **Joshua Agterberg**, Youngser Park, and Carey Priebe, “Concentration bounds on response-based vector embeddings of black-box generative models,” Submitted, 2026.
2. **Joshua Agterberg** and René Vidal, “A High-Dimensional Statistical Theory for Convex and Nonconvex Matrix Sensing,” Submitted, 2026.
3. **Joshua Agterberg**, “Statistical Inference for Linear Functions of Eigenvectors with Small Eigen-gaps,” Submitted, 2026.
4. **Joshua Agterberg**, Minh Tang, and Carey Priebe, “On Two Distinct Sources of Nonidentifiability in Latent Position Random Graph Models,” Submitted, 2020.

CONFERENCE PAPERS (PEER-REVIEWED)

(* denotes students supervised or mentored)

1. Uday Kiran Reddy Tadipatri*, Benjamin D. Haeffele, **Joshua Agterberg**, Ingvar Ziemann, and René Vidal, “Nonconvex Linear System Identification with Minimal State Representation,” *Learning For Dynamics and Control (L4DC)*, 2025.
2. Uday Kiran Reddy Tadipatri*, Benjamin D. Haeffele, **Joshua Agterberg**, and René Vidal, “A Convex Relaxation Approach to Generalization Analysis for Parallel Positively Homogeneous Networks,” *AISTATS*, 2025. (31.3% Acceptance Rate)
3. Liangzu Peng, Juan Elenter, **Joshua Agterberg**, and René Vidal, “TSVD: Bridging Theory and Practice in Continual Learning with Pre-Trained Models,” *ICLR*, 2025. (32.08% Acceptance Rate)
4. **Joshua Agterberg** and Jeremias Sulam, “Entrywise Recovery Guarantees for Sparse PCA via Sparsistent Algorithms,” *AISTATS*, 2022. (29.2% Acceptance Rate)

TALKS

Statistically and Computationally Optimal Estimation and Inference in the Common Subspace Model

- 12/13/2026, CMStatistics, Berlin, Germany (Invited Talk)
- 11/1/2026, Statistical Learning and Data Science Conference, New York, NY (Invited Talk)
- 8/5/2026, Joint Statistical Meetings, Boston, MA
- 6/22/2026, International Symposium on Nonparametric Statistics (ISNPS), Thessaloniki, Greece (Invited Talk)
- 6/8/2026, Statistical Network Analysis and Beyond (SNAB), Hanoi, Vietnam (Invited Talk)
- 1/22/2026, IMSI Workshop on Recent Advances in Random Networks, Chicago, IL (Invited Talk)
- 12/18/2025, International Chinese Statistical Association Conference, Taipei, Taiwan (Invited Talk)
- 12/16/2025, Workshop on Statistical Network Analysis, Imperial College London, England (Invited Talk)

- 12/15/2025, CMStatistics, London, England (Invited Talk)
- 11/17/2025, Statistics Seminar, Iowa State University (Invited Talk)

Statistical Inference for Linear Functions of Eigenvectors with Small Eigengaps

- 4/25/2025, Econometrics Seminar, University of Illinois Urbana-Champaign (Invited Talk)
- 4/3/2025, Statistics Seminar, University of Wisconsin-Madison (Invited Talk)

Estimation and Inference in Tensor Mixed Membership Blockmodels

- 11/16/26, SIAM Conference on the Mathematics of Data Science Minisymposium on Tensor Data Analysis, Salt Lake City, Utah (Invited Talk)
- 6/14/2025, International Indian Statistical Association Conference, Lincoln, Nebraska (Invited Talk)
- 5/8/2025, IMSI Workshop on Tensor Data Analysis and Statistics (Invited Talk)
- 12/16/2024, IMS International Conference on Statistics and Data Science, Nice, France (Invited Talk)
- 11/2/2023, Stochastics Seminar, Georgia Tech (Invited Talk)
- 10/18/2023, Statistical Inference on Networks and High-Dimensional Data Workshop, Brin Mathematics Research Center, University of Maryland College Park (Invited Talk)
- 9/28/2023, Applied Mathematics and Statistics Postdoc Seminar, Johns Hopkins University (Invited Talk)
- 8/21/2023, TRIPODS Postdoc Workshop, Toyota Technology Institute of Chicago

A Precise High-Dimensional Statistical Theory for Convex and Nonconvex Matrix Sensing

- 6/13/2024, Banff International Research Station Workshop on the Mathematics of Deep Learning, Oaxaca, Mexico (Invited Talk)

Statistical Inference for Low-Rank Tensors: Heteroskedasticity, Subgaussianity, and Applications

- 10/19/2024, Statistics Seminar, University of Illinois Urbana-Champaign (Invited Talk)
- 4/6/2024, Statistics Seminar, University of Virginia (Invited Talk)

Estimating Higher-Order Mixed Memberships via the $\ell_{2,\infty}$ Tensor Perturbation Bound

- 2/9/2023, Statistics Department Seminar, University of Illinois-Urbana Champaign (Invited Talk)
- 2/1/2023, Statistics Department Seminar, University of California-Riverside (Invited Talk)
- 1/24/2023, Statistics Department Seminar, University of Virginia (Invited Talk)
- 1/19/2023, Applied Mathematics and Statistics Department Seminar, JHU
- 1/10/2023, University of South Carolina (Invited Talk)
- 12/22/2022, Statistics and Actuarial Science Department Seminar, University of Waterloo (Invited Talk)

Joint Spectral Clustering in Multilayer Degree-Corrected Stochastic Blockmodels

- 12/19/2022, Center for Imaging Science Retreat, JHU
- 10/4/2022, Applied Mathematics and Statistics Student Seminar, JHU
- 8/23/2022, COMPSTAT, Bologna, Italy (Invited Talk)
- 7/11/2022, Statistical Inference for Network Models (NETSCI Satellite), Virtual (Contributed Talk)

Entrywise Estimation of Singular Vectors of Low-Rank Matrices with Heteroskedasticity and Dependence

- 8/11/2022, Joint Statistical Meetings, Washington, D.C. (Contributed Talk)
- 1/20/2022, 2022 TRIPODS Winter School on Interplay between Artificial Intelligence and Dynamical Systems, JHU (Invited Talk)
- 12/6/2021, Venkataraman Lab, JHU (Invited Talk)
- 9/21/2021, Applied Mathematics and Statistics Student Seminar, JHU

Entrywise Recovery Guarantees for Sparse PCA via Sparsistent Algorithms

- 4/19/2022, Applied Mathematics and Statistics Student Seminar, JHU
- 3/30/2022, AISTATS, 2022, ~~Valencia, Spain~~ Virtual

From RDPGs to General Signal Plus Noise Models

- Guest Lecture for 553.742 Statistical Inference on Graphs, JHU

Nonparametric Two-Sample Hypothesis Testing for Random Graphs with Negative and Repeated Eigenvalues

- 8/10/2021, Joint Statistical Meetings, ~~Seattle, WA~~ Virtual (Topic-Contributed Talk)
(**Received Best Presentation Award in the Nonparametric Statistics Student Competition**)
- 2/2/2021, Applied Mathematics and Statistics Student Seminar, JHU
- 8/6/2020, Joint Statistical Meetings, ~~Philadelphia, PA~~ Virtual (Contributed Talk)
- 4/7/2020, MINDS Seminar, JHU

On Two Distinct Sources of Nonidentifiability in Latent Position Random Graph Models

- 1/28/2020, Applied Mathematics and Statistics Student Seminar, JHU

Vertex Nomination, Consistent Estimation, Adversarial Modification

- 4/23/2019, Applied Mathematics and Statistics Student Seminar, JHU

TEACHING

University of Illinois Urbana-Champaign

Fall 2026	Instructor , Statistics 511, Advanced Mathematical Statistics
Fall 2025	Instructor , Statistics 578, Topics in Statistics PhD-level course on spectral methods, network analysis, and nonconvex optimization
Fall 2025	Instructor , Statistics 511, Advanced Mathematical Statistics PhD-level required course on mathematical statistics
Fall 2024	Instructor , Statistics 410, Probability and Statistics II Upper level undergraduate and master's-level course on calculus-based probability and statistics

Johns Hopkins University

Fall 2022	Instructor , 500.111: Statistics and Data Science with Networks Undergraduate course for first and second-year engineering students on basics of data science and network analysis
Winter 2021	Instructor , 553.283 Introduction to R Undergraduate introduction to statistical computing with R

Fall 2021	Instructor , 500.111: Statistics and Data Science with Networks
Summer 2021	Instructor , Master's Program Statistics Review Review sessions on mathematical statistics for Master's students
Winter 2021	Instructor , 553.283 Introduction to R
Summer 2020	Instructor , Master's Program Statistics Review
Summer 2020	Teaching Assistant , 553.310 Probability and Statistics for Vittorio Loprinzo Undergraduate introduction to statistics
Fall 2019	Co-created a topics course in probability with another graduate student
Summer 2019	Instructor , Master's Program Statistics Review
Summer 2019	Teaching Assistant , 553.310 Probability and Statistics for Vittorio Loprinzo
Spring 2019	Teaching Assistant , 553.762 Nonlinear Optimization II for Professor Daniel Robinson Graduate-level course on constrained optimization
Fall 2018	Teaching Assistant , 553.730 Statistical Theory for Professor Carey Priebe Graduate-level course on statistical theory
Summer 2018	Instructor , Master's Program Statistics Review

University of Wisconsin-Madison (Wisconsin School of Business, Risk and Insurance)

Spring 2017	Grader , ActSci 655 Health Analytics for Professor Margie Rosenberg Upper-level undergraduate course on statistical analysis for health insurance
Fall 2016	Grader , ActSci 651 Life Contingencies II for Professor Paul Johnson Upper-level undergraduate course on actuarial mathematics for life insurance
Spring 2016	Grader , ActSci 650 Life Contingencies I for Professor Margie Rosenberg Upper-level undergraduate course on actuarial mathematics for life insurance

University of Wisconsin-Madison (School of Music)

2015-2017	Private Piano Instructor
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STUDENTS SUPERVISED OR MENTORED

- Georgios Avdis (Statistics PhD, UIUC), 2024-Present.
- Zhexu (Alex) Jin (Statistics PhD, UIUC), 2025-Present.
- Rajdeep Brahma (Statistics PhD, UIUC), 2024-Present (Co-advised with Yuguo Chen).
- Yanlin Wang (Statistics Master's, UIUC), 2024-2025.
- Uday Kiran Reddy Tadipatri (ESE PhD, University of Pennsylvania), 2023-2025.

PhD Committee: ByeongJip Kim (UIUC Statistics, 2025)

PROFESSIONAL ACTIVITIES AND SERVICE

- Reviewer for the following journals and conferences: Journal of the Royal Statistical Society, Series B (1), Annals of Statistics (3), Electronic Journal of Statistics (3), Journal of the American Statistical Association (4), Journal of Computational and Graphical Statistics (5), IEEE Transactions on Information Theory (2), SIAM Journal on Matrix Analysis and Its Applications (1), IEEE Signal Processing Letters (3), Biometrika (1), Journal of Business and Economic Statistics (2), Journal of the Korean Statistical Society (2), IEEE Transactions on Signal Processing (2), AISTATS (4), Statistics and its Interface (1), IEEE Transactions on Pattern Analysis and Machine Intelligence (1), Statistics in Medicine (2), Journal of Statistical Planning and Inference (3),

Journal of Multivariate Analysis (1), European Journal of Applied Mathematics (2), DeepMath (1)

- Committee Membership at UIUC: PhD Student Committee (2024-2026), Qualifying Exam Committee (2025-2026), Seminar/Colloquium Committee (2025-2026)
- Chair of 2022 Joint Statistical Meetings session on Network Data Analysis
- Member of the ASA, IMS

HONORS AND AWARDS

Johns Hopkins University

Summer 2022	Acheson J. Duncan Fund for the Advancement of Research in Statistics Travel Award (awarded twice)
Spring 2022	MINDS (Mathematical Institute of Data Science) Fellowship
Summer 2021	Best Presentation Award, JSM Student Competition in Nonparametric Statistics
Spring 2021	IMS (Institute of Mathematical Statistics) Hannan Graduate Student Travel Award
Spring 2021	MINDS (Mathematical Institute of Data Science) Fellowship
Spring 2021	Finalist for JSM Student Competition in Nonparametric Statistics
2021-2023	Applied Mathematics and Statistics Teaching Fellow
2020-2021	Applied Mathematics and Statistics Apprentice Teaching Fellow
2019-2023	Charles and Catherine Counselman Fellowship
Spring 2020	MINDS (Mathematical Institute of Data Science) Fellowship

University of Wisconsin-Madison

Spring 2017	Graduated with distinction
Spring 2017	DW Simpson Scholarship
Fall 2016	Bicknell Scholarship
2013-2014	Arthur C. Nielsen Scholarship
2013	Directly Admitted to Wisconsin School of Business
2014-2017	Dean's list

ADDITIONAL EXPERIENCE

Summer 2017	Analytics Intern , CNA Financial, Chicago, IL
Summer 2016	Actuarial Intern , CNA Financial, Chicago, IL
Summer 2015	Actuarial Intern , CUNA Mutual Group, Madison, WI

ADDITIONAL TECHNICAL QUALIFICATIONS

Passed three actuarial exams (P, FM, and MFE)

HOBBIES

Trail and ultra running, cycling, lifting weights, reading fantasy